CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 11: Geometric Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The general term

1. 1701 is the nth term of the G.P. $7, 21, 63, \ldots$ Find (a) n (b) the sum of these n terms. ICSE 2025, 3 marks

SOLUTION

(a) $a = 7$, $r = 3$. $7 \times 3^{n-1} = 1701$, so $3^{n-1} = 243 = 3^5$ and $n = 6$.

(b) $S_6 = \dfrac{a(r^6 - 1)}{r - 1} = \dfrac{7(729 - 1)}{2} = 7 \times 364$

Answer: (a) $n = 6$ **(b) sum** $= 2548$

2. The 4th, 6th and last terms of a G.P. are 10, 40 and 640. The common ratio is positive. Find the first term, the common ratio and the number of terms. ICSE 2020, 4 marks

SOLUTION

$ar^3 = 10$ and $ar^5 = 40$. Dividing: $r^2 = 4$, so $r = 2$.

$a = \dfrac{10}{8} = \dfrac{5}{4}$

Last term: $\dfrac{5}{4} \times 2^{n-1} = 640$, so $2^{n-1} = 512 = 2^9$ and $n = 10$.

Answer: $a = \dfrac{5}{4}$, $r = 2$, $n = 10$

3. The 4th term of a G.P. is 16 and the 7th term is 128. Find the first term and the common ratio. ICSE 2018, 3 marks

SOLUTION

$ar^3 = 16$ and $ar^6 = 128$. Dividing: $r^3 = 8$, so $r = 2$.

$a = \dfrac{16}{8} = 2$

Answer: first term 2, common ratio 2

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2 Sum of n terms

4. $15, 30, 60, 120, \ldots$ is a G.P. (a) Write its nth term. (b) How many terms add up to 945? ICSE 2024, 4 marks

SOLUTION

(a) $a = 15$ and $r = 2$, so $T_n = 15 \times 2^{n-1}$.

(b) $S_n = \dfrac{a(r^n - 1)}{r - 1} = 15(2^n - 1) = 945$

$2^n - 1 = 63$, so $2^n = 64 = 2^6$.

Answer: (a) $T_n = 15 \times 2^{n-1}$ (b) 6 terms

5. A G.P. has first term 3, last term 96 and common ratio 2. Find (a) the number of terms (b) their sum. ICSE 2019, 3 marks

SOLUTION

(a) $3 \times 2^{n-1} = 96$, so $2^{n-1} = 32 = 2^5$ and $n = 6$.

(b) $S_6 = \dfrac{3(2^6 - 1)}{2 - 1} = 3 \times 63$

Answer: (a) 6 terms (b) 189

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